

RESEARCH ARTICLE

Composition, abundance and distribution of rodent species in Alemsaga Priority State Forest and farmlands, northwestern Ethiopia

Bewketu Takele¹  | Mesele Yihune²  | Afework Bekele²

¹Department of Biology, Debre Markos University, Debre Markos, Ethiopia

²Department of Zoological Sciences, Addis Ababa University, Addis Ababa, Ethiopia

Correspondence

Mesele Yihune, Department of Zoological Sciences, Addis Ababa University, P.O. Box 1176, Addis Ababa, Ethiopia.
Email: mesyih@gmail.com

Funding information

Addis Ababa University

Abstract

The aim of this study was to investigate species composition, relative abundance, habitat preference and seasonal variations of rodents in Alemsaga Priority State Forest and the nearby farmlands. A total of 1311 individual rodents were captured in six permanent grids and line transects, comprising of nine species, namely *Lophuromys simensis*, *Arvicanthis abyssinicus*, *Desmomys harringtoni*, *Mastomys natalensis*, *Stenocephalemys albipes*, *Rattus rattus*, *Arvicanthis dembeensis*, *Mus musculus* and *Acomys cahirinus*. *Hystrix cristata* and *Tachyoryctes splendens* were also recorded but not captured in trapes. About 44.4% of captured species are endemic to Ethiopia. *Lophuromys simensis* was captured from all habitats, while *Acomys cahirinus* was recorded only in Carissa land. The highest abundance of rodents was recorded in crop fields and the lowest in Carissa land. The overall trap success was 14.76%. Statistical variation was observed between individual rodent species among different habitats. Highest rodent populations were recorded during the wet season. Females comprised more individuals than males. Adults outnumbered sub-adults and juveniles. Vegetation composition, altitudinal variations and human disturbance were the major factors affecting the distribution of rodents in the area. Attention should be given by the local administrators to conserve the area.

KEYWORDS

Alemsaga Forest, distribution, diversity, Ethiopia, rodents, species composition

Résumé

Le but de cette étude était d'étudier la composition des espèces, l'abondance relative, les préférences en matière d'habitat et les variations saisonnières des rongeurs dans la forêt domaniale d'intervention prioritaire d'Alemsaga et les terres agricoles adjacentes. Au total, 1 311 rongeurs individuels de neuf espèces, à savoir *Lophuromys simensis*, *Arvicanthis abyssinicus*, *Desmomys harringtoni*, *Mastomys natalensis*, *Stenocephalemys albipes*, *Rattus rattus*, *Arvicanthis dembeensis*, *Mus musculus* et *Acomys cahirinus* ont été capturés sur six grilles et transects linéaires. Les espèces *Hystrix cristata* et *Tachyoryctes splendens* ont également été signalées mais n'ont pas été capturées dans les pièges. Environ 44,4% des espèces capturées sont endémiques en Éthiopie. *Lophuromys simensis* a été capturé dans tous les habitats, tandis qu'*Acomys*

cahrinus n'a été signalée que dans les terres de Carissa. La plus forte abondance de rongeurs a été enregistrée dans les champs cultivés et la plus faible dans les terres de Carissa. Le succès global du piège était de 14,76 %. Une variation statistique a été observée entre différentes espèces de rongeurs dans différents habitats. Les populations de rongeurs les plus importantes ont été constatées pendant la saison des pluies. Le nombre d'individus femelles était plus élevé que celui des individus mâles. Les individus adultes étaient plus nombreux que les individus non-adultes et juvéniles. La composition de la végétation, les variations d'altitude et les perturbations humaines étaient les principaux facteurs affectant la répartition des rongeurs dans la région. Une attention doit être accordée à ces résultats par les administrateurs locaux aux fins de conservation de la région.

1 | INTRODUCTION

Rodents are highly diversified in their mode of life, morphology, ecology and behaviour (Lavrenchenko & Bekele, 2017). Their diversity ranges from tiny pigmy mice to large-sized capybaras, from arboreal flying squirrels to subterranean mole rats or semi-aquatic species like beavers and water voles (Vaughan et al., 2000). The variation in body weight also ranges from 5 g to 50 kg (Vaughan et al., 2000). In Ethiopia, rodents are more adapted to the Afro-alpine environment than large mammals due to their ability to avoid temperature fluctuations by burrowing (Yihune & Bekele, 2012).

Several studies have been conducted on rodent species richness and distribution in Ethiopia including Chilalo-Galama Mountain Ranges (Kasso et al., 2010), the Simien Mountains National Park (Yihune & Bekele, 2012), Kaka and Hunkolo, Southeast Ethiopia (Girma et al., 2012), Chebera Churchura National Park (Datiko & Bekele, 2013), Awi Zone (Bantihun & Bekele, 2015), Bale Mountains National Park (Fekadu et al., 2015) and Choke Mountain (Simeneh, 2016). Most previous studies on rodent populations targeted the protected southwestern forests, south and southeastern highlands

(Chekol et al., 2012; Habtamu & Bekele, 2008; Kasso & Bekele, 2011). However, there is no previous study on rodents species composition, abundance and distribution in Alemsaga Priority State Forest located in the northwest highland of Ethiopia.

The ever-increasing demand for agricultural land in Ethiopian highlands resulted in increased intensity of livestock grazing and cultivation on marginal lands. Alemsaga Priority State Forest is part of the northwest highlands of Ethiopia. Therefore, this study aimed at filling the existing knowledge gaps in this part of Ethiopia.

2 | MATERIALS AND METHODS

2.1 | Description of the study area

Alemsaga Priority State Forest is located in Amhara Regional State, South Gondar Zone, northwestern Ethiopia at a distance of 650 km from Addis Ababa and 16 km from Debre Tabor Town (Figure 1). It is located between 11°54'–11°56'N latitude and

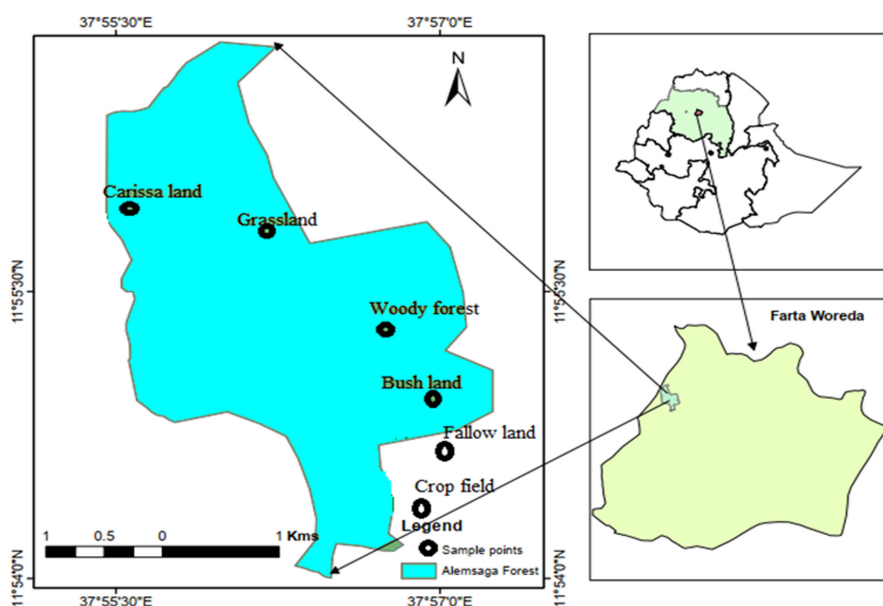


FIGURE 1 Map of the study area and trapping sites

037°55′–037°57′E longitude (ORDA, 2012). Its elevation ranges from 2100 to 2600 m a.s.l. The total area of the forest is about 880 ha (ORDA, 2010).

The area is rich in animal and plant diversity. Grivet monkey (*Cercopithecus aethiops*), Olive baboon (*Papio anubis*), Aardvark (*Orycteropus afer*), Spotted hyaena (*Crocuta crocuta*), African civet (*Civettictis civetta*), Ethiopian Dwarf Mongoose (*Helogale hirtula*), golden jackal (*Canis aureus*), Common duiker (*Sylvicapra grimmia*), Bush pig (*Potamochoerus larvatus*), Leopard (*Panthera pardus*) and hyrax are the common species in the forest (ORDA, 2010). Furthermore, birds like Wattled ibis (*Bostrychia carunculata*), Harwood's francolin (*Francolinus harwoodi*), Black winged Lovebird (*Agapornis taranta*), Abyssinian woodpecker (*Dendropicus abyssinicus*), Abyssinian longclaw (*Macronyx flavicollis*), Rouget's rail (*Rougetius rougetii*), Banded barbet (*Lybius undatus*) and Helmeted guinea fowl (*Numida meleagris*) are residing in the forest (ORDA, 2010). About 66 woody plant species are reported in the area by Esubalew et al. (2019).

The study was conducted from August 2018 to September 2020. Before the actual data collection, a preliminary survey was conducted. During the survey, habitats were classified based on the dominant plant species, altitude and disturbance level. Six representative habitats (Carissa land, grassland, woody forest, bushland, fallow land and crop field) were selected for the study.

A permanent live trapping grid of 70 × 70 m (4900 m²) was established in each habitat (Yihune & Bekele, 2012). A total of 49 Sherman live traps (5.5*6.5*16 cm) were used per grid set in trapping stations at 10-m intervals. The six live trapping grids were set at more than 500 m between any two grids to ensure independent captures (Aplin et al., 2003). Trap stations were marked with yellow plastic tags (Geleta, 2010) to identify the position of the traps during inspection and collection. Traps were set for five consecutive days and nights for each habitat during trapping session. Traps were baited with peanut butter and barley flour as well as bread (Kasso et al., 2010; Yihune & Bekele, 2012).

Traps were covered with vegetation (grasses and leaves) to protect captured animals from harsh environmental conditions and to minimise visibility to predators. Traps were checked twice a day, early in the morning (6:30 to 8:30 a.m.) and in the late afternoon (16:30 to 18:30 p.m.). During inspection, traps were cleaned and rebaited where necessary due to loss of the bait to other animals or insects. Captured rodents were removed from the traps, placed in a transparent polyethylene bag and weighed to the nearest gram using 100 g Pesola spring balance. Each captured animal was identified to species level, marked by toe clipping and released back at the site where it was captured (Aplin et al., 2003; Kingdon et al., 2013). Sex and approximate age were recorded based on weight and coat colour. Furthermore, reproductive conditions including imperforate or perforated vagina, size of nipples, distended abdomen and body weight for females and the position and size of testes for males were also recorded following Bekele (1996a) and Mahlaba and Perrin (2003). For juvenile sex identification, the distance between the genital papilla and anus was also used (Aplin et al., 2003).

For accurate body measurement and identification, 25 snap traps were used in each permanent grid for five consecutive days and nights. Snap traps were spaced at about 10-m interval along five transect lines in all habitat types. The transect lines were set at a distance of 200 m from the live trapping grids to avoid home range effect. Animals captured in snap traps were identified to species level (Vieira, 2003). For morphological body measurements, only adult rodents were used. The skins and skulls of representative snap-trapped rodents were prepared and brought to the Zoological National History Museum of Addis Ababa University, where final identification of specimens was confirmed.

All the collected data were analysed using SPSS version 21. Shannon–Wiener diversity indices were used to analyse the diversity of species for different habitats (Shanon & Weaver, 1949). One-way ANOVA was used to show population variations of each species of rodents in the study area. Chi-square test was used to test for significant differences of rodent species between seasons and sexes. Effective trap nights, percentage of trap success and relative abundances were also calculated. Finally, similarity indices between habitats were examined.

3 | RESULTS

A total of 1311 individual rodents were captured in 8880 trap nights in six different habitats. They belonged to 3 families and 11 species. Of these, nine species were from the family Muridae (*Lophuromys simensis*, *Arvicanthis abyssinicus*, *Desmomys harringtoni*, *Mastomys natalensis*, *Stenocephalemys albipes*, *Rattus rattus*, *Arvicanthis dembeensis*, *Mus musculus* and *Acomys cahirinus*). The other two were from the family Spalacidae (*Tachyorctes splendens*) and Hystricidae (*Hystrix cristata*). *Hystrix cristata* and *T. splendens* were not captured in either live or snap traps. Their occurrence in the study area was confirmed through direct observation and/or indirect evidence. There was no statistically significant variation in the species range of distribution in the six habitats ($\chi^2 = 0.001$, $df = 40$, $p > 0.05$).

Each species in each habitat was categorised as rare (0.01–5%), uncommon (5.01–10%), common (10.01–20%) and abundant (>20%) (Table 1).

Of the total rodents captured, 1140 were live trapped and the remaining 171 snap-trapped. In all sessions of the study periods, 1155 individuals (including snap-trapped rodents) were new captures and the remaining 156 were recaptures. The overall percentage of trap success rate during the study period was 14.76%. It was higher in the live traps (19.39%) compared to the snap traps (5.7%) (Table 2). The null hypothesis that supposed all rodent species have equal number of individuals in the study area was not accepted as the relative abundance of individuals of different species shows significant variations ($p < 0.005$).

The total number of captures in the six habitats of the study area showed statistically significant variations ($\chi^2 = 29.51$, $df = 5$, $p < 0.05$). Similarly, rodent trap success rate showed variations in different habitats (Table 3).

TABLE 1 Distributional patterns of rodent species in different habitats and altitudes (+ = rare, ++ = uncommon, +++ = common, ++++ = abundant, – = not recorded and * observed)

Species	Study habitats (altitudes)					
	CL (2109 m asl)	GL (2228 m asl)	WF (2350 m asl)	BL (2440 m al.)	FL (2470 m asl)	CF (2560 m asl)
<i>Lophuromys simensis</i>	++++	++++	++++	++++	++++	+++
<i>Arvicanthis abyssinicus</i>	–	–	–	–	++++	++++
<i>Desmomys harringtoni</i>	++++	+++	+++	++++	+	–
<i>Mastomys natalensis</i>		+++	+++	++++	+++	+++
<i>Stenocephalemys albipes</i>	+	++++	++++	+++	–	–
<i>Rattus rattus</i>	+	++++	–	–	–	–
<i>Arvicanthis dembeensis</i>	–	–	–	–	++	+++
<i>Mus musculus</i>	–	+	–	+	++	+
<i>Acomys cahirinus</i>	+	–	–	–	–	–
<i>Tachyoryctes splendens</i>	–	–	–	*	*	*
<i>Histrix cristata</i>	*	*	*	*	*	*

Abbreviations: BL, bushland; CF, crop field; CL, Carissa land; FL, fallow land; GL, grassland; WF, woody forest.

TABLE 2 Relative abundance of rodent species in study area

Species	Total catch	Relative abundance (%)
<i>Lophuromys simensis</i>	407	31.05
<i>Arvicanthis abyssinicus</i>	252	19.22
<i>Desmomys harringtoni</i>	205	15.6
<i>Mastomys natalensis</i>	168	12.8
<i>Stenocephalemys albipes</i>	138	10.53
<i>Rattus rattus</i>	55	4.2
<i>Arvicanthis dembeensis</i>	50	3.8
<i>Mus musculus</i>	31	2.4
<i>Acomys cahirinus</i>	5	0.38
Total	1311	100

The overall trap success in relation to altitudinal ranges is shown in Table 4. The highest trap success was at altitudinal range between 2447 and 2560 m a.s.l. (16.62%) followed by the altitudinal range between 2350 and 2440 m a.s.l. (14.19%). The lowest trap success was recorded at altitudinal range between 2109 and 2228 m a.s.l. (13.48%).

Habitat preference of rodent species in the study area is given in Figure 2. Statistically significant variations were observed ($\chi^2 = 1177.03$, $df = 40$, $p < 0.05$) in the abundance of each rodent species in various habitats of the study area.

Seasonal variations in abundance of different rodent species and trap success are presented in Table 5. The total catch of both live and snap-trapped rodents between the wet and dry seasons was 761 (58.05%) and 550 (41.95%), respectively. Total captures showed statistically significant variation between seasons ($\chi^2 = 34.43$, $df = 1$, $p < 0.05$).

TABLE 3 Trap success of rodents in different habitats of the study area

Habitats	Total capture	Trap nights	Trap success (%)
CL (2109 m asl)	161	1480	10.88
GL (2228 m asl)	238	1480	16.08
WF (2350 m asl)	207	1480	13.99
BL (2440 m asl)	213	1480	14.49
FL (2470 m asl)	223	1480	15.10
CF (2560 m asl)	269	1480	18.18
Total	1311	8880	14.76

Abbreviations: BL, bushland; CF, crop field; CL, Carissa land; FL, fallow land; GL, grassland; WF, woody forest.

Of the habitat types, Carissa land, bushland and crop field constituted five rodent species during the study period. The forest had the least species richness (four rodent species).

The highest diversity index was recorded in the grassland habitat ($H' = 1.57$) followed by the bushland area ($H' = 1.42$). Simpson's diversity index was also highest in the Carissa land ($D = 0.401$) followed by fallow land ($D = 0.34$). The highest species evenness was recorded in the woody forest habitat ($E = 0.8971$) followed by bushland ($E = 0.8293$) (Table 6). In general, the forest habitat showed relatively higher diversity index than the nearby farmlands during the study period.

Out of the total captures, males and females comprised 592 and 719 individuals respectively. Statistically significant variations were observed in the abundance of male and female animals between seasons ($\chi^2 = 33.96$, $df = 1$, $p < 0.05$). The ratio of females to males was 1.0:1.2. From the total captures, adults comprised 657 individuals, sub-adults 461 and juveniles 193. The difference in age groups was

TABLE 4 Trap success for different rodent species across altitudinal ranges

Species	Trap success and altitudinal ranges		
	2109–2228 (m asl)	2350–2440 (m asl)	2447–2560 (m asl)
<i>Lophuromys simensis</i>	4.66	5.81	3.33
<i>Arvicanthis abyssinicus</i>	0	0	8.51
<i>Desmomys harringtoni</i>	4.02	2.84	0.1
<i>Mastomys natalensis</i>	0.81	2.53	2.3
<i>Stenocephalemys albipes</i>	1.93	2.74	0
<i>Rattus rattus</i>	4.12	0	0
<i>Arvicanthis dembeensis</i>	0	0	1.69
<i>Mus musculus</i>	0.03	0.27	0.74
<i>Acomys cahrinus</i>	0.17	0	0
Total individuals	399	420	492
Trap night	2960	2960	2960
Trap success	13.48	14.19	16.62

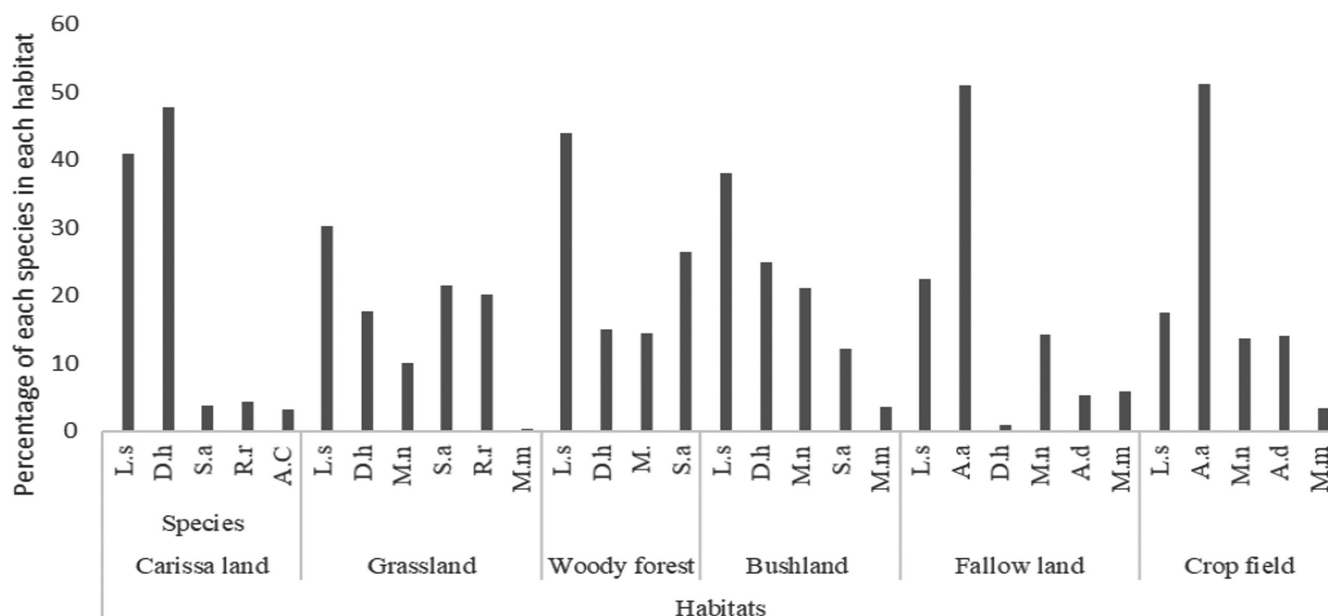


FIGURE 2 Habitat preference of recorded rodents species in the study area (L.l = *L. simensis*, A.b = *A. abyssinicus*, D.h = *D. harringtoni*, M.n = *M. natalensis*, S.a = *S. albipes*, R.r = *R. rattus*, A.d = *A. dembeensis*, M.m = *M. musculus* and A.c = *A. cahrinus*)

statistically significant ($\chi^2 = 35.96$, $df = 2$, $p < 0.05$) between seasons (Table 7).

4 | DISCUSSION

The study revealed occurrence of 11 rodent species which comprises 12% of the Ethiopian rodent species currently described. The results agree with the findings of Fekadu et al. (2015) and Simeneh (2016) who reported about 12% of the country's rodent species in the Bale Mountains National Park and 10% in Choke Mountain, East Gojjam Zone. Four of the captured species are endemic to Ethiopia. These comprise about 11.11% rodent species endemic to Ethiopia. Nearly, the same number of endemic rodent species was reported

in the Choke Mountain by Simeneh (2016) and Kostin et al. (2020). However, 60% of Ethiopian endemic rodent species were reported from Chilalo–Galama Mountain range, Ethiopia (Kasso et al., 2010). Nearly, the same numbers of species were reported from Simien Mountains National Park (Craig et al., 2020), Borena Sayint National Park (Moges et al., 2016) and around Asella Town (Kasso & Bekele, 2017).

However, Fekadu et al. (2015), Admas and Yihune (2016), Simeneh (2016) and Asefa (2018) reported lower numbers of species than in the present study in several other habitats in Ethiopia. On the other side, Workineh et al. (2011); Dubale and Ejigu (2015); Workineh and Reddy (2016); Kasso and Bekele (2017) and Dinaw et al. (2017) reported high number of species in the country. These variations might be associated with availability of resources, patchiness in the

TABLE 5 Seasonal variations in trap success of rodent species

Species	Season	Trapped rodents	Total trap nights	Trap success (%)	χ^2 value (df)	p Value
<i>Lophuromys simensis</i>	Wet	229	4440	5.16	6.644 (1)	0.01
	Dry	178	4440	4.01		
<i>Arvicanthis abyssinicus</i>	Wet	163	4440	3.67	22.256 (1)	0.00
	Dry	89	4440	2.00		
<i>Desmomys harringtoni</i>	Wet	127	4440	2.86	11.942 (1)	0.001
	Dry	78	4440	1.76		
<i>Mastomys natalensis</i>	Wet	84	4440	1.89	0.00 (1)	1.00
	Dry	84	4440	1.89		
<i>Stemocephalomys albipes</i>	Wet	85	4440	1.91	7.518 (1)	0.006
	Dry	53	4440	1.19		
<i>Rattus rattus</i>	Wet	36	4440	0.81	5.252 (1)	0.022
	Dry	19	4440	0.43		
<i>Arvicanthis dembeensis</i>	Wet	26	4440	0.59	0.080 (1)	0.779
	Dry	24	4440	0.54		
<i>Mus musculus</i>	Wet	10	4440	0.23	3.915 (1)	0.048
	Dry	21	4440	0.47		
<i>Acomys cahirinus</i>	Wet	1	4440	0.02	0.080 (1)	0.777
	Dry	4	4440	0.10		

Habitats	Altitude (m a.s.l.)	Species richness	Total capture	H'	D	Hmax	E
CL	2109	5	161	1.09	0.401	1.61	0.5919
GL	2228	6	238	1.57	0.2194	1.79	0.8052
WF	2350	4	207	1.28	0.3073	1.39	0.8971
BL	2440	5	213	1.42	0.2675	1.61	0.8293
FL	2470	6	223	1.32	0.3386	1.79	0.6252
CF	2560	5	269	1.31	0.3337		0.7414

TABLE 6 Species diversity index values of various habitats of the study area (H' = Shannon-Weaver diversity index; D = Simpson's diversity index (1-D); Hmax = Natural logarithm of the total number of species; E = Evenness)

TABLE 7 Age class distribution of rodents and relative abundance of each age group between seasons

Age class	Wet season			Dry season		
	Total capture	Trap night	Trap success (%)	Total capture	Trap night	Trap success (%)
Adult	345	4440	7.8	312	4440	7.03
Sub-adult	268	4440	6.0	193	4440	4.3
Juvenile	148	4440	3.3	45	4440	1.0

distribution of resources caused by habitat fragmentation, density of vegetation, habitat disturbance, climatic conditions, disease, behavioural and ecological adaptability and habitat utilisation (Admas & Yihune, 2016; Kasso & Bekele, 2017). The rodent species recorded in the present study are similar to those reported in northwestern highlands and central Ethiopia (Yihune and Bekele (2012), Simenah (2016), Craig et al. (2020) and Kostin et al. (2020).

The most abundant and widely distributed species in this study was *L. simensis* (Simien brush-furred rat). The genus *Lophuromys* is widely distributed in Ethiopian highlands (Bryja et al., 2018). Among

these, *L. simensis* and *L. menageshae* were reported in northern parts of the country (Bryja et al., 2018; Craig et al., 2020; Kostin et al., 2020; Lavrenchenko & Bekele, 2017). *Lophuromys simensis* is distributed around Lake Tana, Simien Mountains National Park, Borena Saynt National Park and Debre Tabor (Bryja et al., 2018; Craig et al., 2020; Lavrenchenko et al., 2004; Moges et al., 2016). *Arvicanthis abyssinicus* (grass rat) was the second in abundance, but its distribution was limited in fallow land and crop fields.

Though not captured, the presence of *A. abyssinicus* was expected in grassland and bushland habitats shown by observations of

nests constructed from grasses. Bekele and Yalden (2013) reported that this species was the most abundant in grassland and agricultural fields. Similarly, a recent report by Kostin et al. (2020) revealed high abundance of grass rats at high elevation plateaus in the north-western part of Ethiopian highlands (Choke Mountain). *Arvicanthis abyssinicus* was also reported as a major pest and was abundant in the grassland and moorland habitats in northern Ethiopia (Happold, 2013). Yihunie and Bekele (2012) also reported high numbers and extended range of distribution in the Simien Mountains National Park. This habitat association of the grass rats might be related to the high biotic potential and presence of beneficial food items in farmland areas. The presence of drought resistant crops (*Lathyrus sativus* and *Cicer arietinum*) during the dry season in most farmland areas might have influenced its distribution in our study area. The recently implemented strategy by the regional government in the study area, 'free farmland from livestock after harvesting for soil conservation', might have also contributed to the high occurrence of this species in the present study area.

Desmomys harringtoni (Harrington's rat) was the third most abundance species recorded in all habitats except farmlands. This species is most common in the Ethiopian highlands at altitudes between 1800 and 2800 m a.s.l. (Lavrenchenko & Bekele, 2017). Similarly, Kassa and Bekele (2008) reported high abundance of this endemic rodent species over a large area. Habtamu and Bekele (2012) also reported similar distributional range of this species in Jiren Mountain, Jimma. The very low relative abundance of this endemic species in the nearby farmland indicated that this was not a favourable habitat for the species.

Mastomys natalensis (multimammate rat) was also recorded at high altitudinal ranges from 2228 to 2560 m a.s.l. In southern Ethiopia, Bekele and Yalden (2013) reported its distribution at lower altitude ranging from 430 to 4400 m a.s.l. This pattern of distribution shows that the species is adapted to diverse ecological situations. Different studies in various parts of Ethiopia (Datiko & Bekele, 2013; Kostin et al., 2020) also reported the generalist behaviour of this species. A widespread distribution of *M. natalensis* is also reported in Ethiopia, on the northwestern plateau and south of the Ethiopian Rift Valley (Kostin et al., 2020).

In the present investigation, *S. albipes* was recorded from 2009 to 2470 m a.s.l. In Ethiopia, this endemic species has been reported at different altitudinal ranges between 528 m a.s.l. (Habtamu & Bekele, 2008) and 3300 m a.s.l. (Bekele, 1996b). These results were similar to those reported by Workineh and Reddy (2016) in Nech Sar National Park with high population in grassland and bushland habitat. Yalden and Largen (1992) reported *S. albipes* as a very common rat of Ethiopian plateau associated with natural vegetation. Bryja et al. (2018) revealed *S. albipes* is widely distributed throughout the Ethiopian highlands. Kostin et al. (2020) also reported this species to be widespread in the northern part of the Ethiopian highlands including the top of Choke Mountain.

Rodent species diversity and relative abundance showed variations among habitats. Human interference, rainfall pattern and vegetation cover were the main factors influencing the abundance and

distribution of rodents. Resource accessibility, habitat complexity and elevation appeared to regulate rodent diversity and abundance in our study area. According to Lavrenchenko and Bekele (2017), habitat usage of small mammals is linked to ecological variables, biological interactions and species necessities. Grassland and fallow land habitats in this study accounted for the highest number of species (6), while the forest had the lowest (4). These results conform with studies reported by Getachew et al. (2016) who found higher numbers of rodent species in natural habitats than farmland. Unlike in our study, Admas and Yihune (2016) recorded high numbers of rodent species in farmland of East Gojjam zone. The low numbers of rodent species in woody forest habitat might be associated with specialist species for this type of habitat. This may also be the case for other species including *A. abyssinicus* and *A. dembeensis* in fallow land and crop fields as well as *A. cahirinus* in *Carissa* land. The relatively low diversity index in the nearby farmland than the forest in this study indicated the unfavourable conditions of these habitats as a result of human disturbance.

The higher abundance of rodents in a crop fields might be associated with better resources during the wet season favouring the pest rodent species like *Arvicanthis* spp. (Dubale & Ejigu, 2015). Farmland might provide good cover and preferred food items during fruiting and maturation stages. An investigation carried out in and around Katavi National Park, Tanzania, also revealed higher rodent species richness and abundance outside the Park than inside the Park (Caro, 2001). In this study, grassland recorded more species of rodents. Geleta (2010) also reported high rodent abundance in grassland sites in Komto Protected Forest, Western Ethiopia. Dinaw et al. (2017) reported similar findings in the Chebera Churchura National Park. The grasslands in this study area are well protected from grazers throughout the year. In addition, grasslands might be favoured by most of the rodent species because of good cover. Furthermore, fresh grasses could be nutritionally preferable for rodents particularly at the beginning of the wet season (Auffray et al., 2009). Relatively low numbers of rodents were recorded in the *Carissa* land. This could be associated with unsuitability for free movement due to dry spines in the *Carissa* shrubs. Plant diversity and metabolites affect feeding habits of rodents as these metabolites determine the nutritive quality and digestibility (Auffray et al., 2009).

The abundance of rodents was higher during the wet season than the dry season. The possible reason for higher populations during the wet season might be due to an increase in reproduction rates, availability of food and shelter. Rain initiates the growth of fresh plants, especially grasses like *Cynodon aethiopicus*. During the dry season, rodents can migrate to the nearby habitats in search of wet and succulent plants to supplement their water need. Datiko et al. (2007) stated that reduced ground cover and food shortage might be the main reasons for the decrease in rodent populations during the dry season. More female rodents were recorded than males. This might be associated with males being mobile, spending more time and energy in search of females for mating and exposure to predators (Moges et al., 2016).

In this study, high numbers of adults and sub-adults than juveniles were recorded. This might be due to adults and sub-adult

rodents being more active than juveniles. However, Admas and Yihune (2016) recorded more sub-adults than adults from Yekoche Forest, East Gojjam, Ethiopia. High numbers of juveniles were recorded during the wet season. This could be associated with breeding of most rodents during the wet season. The composition and abundance of rodents in the study area seemed to be influenced by vegetation and altitude. On the contrary, human encroachment in natural habitats, illegal cutting of trees for fuel and construction, night-time livestock grazing and poaching are the main disturbances in the study area. These activities could influence species richness, diversity and abundance of rodent species. Therefore, there is a need for measures to be observed to conserve biodiversity of fauna and flora to reduce the negative impact on rodent species in the area.

ACKNOWLEDGEMENT

We would like to thank Addis Ababa University and Debre Markos University for financial support.

CONFLICT OF INTEREST

The authors have no conflict of interest in relation to this work.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Bewketu Takele  <https://orcid.org/0000-0002-7052-2301>

Mesele Yihune  <https://orcid.org/0000-0001-7245-9177>

REFERENCES

- Admas, A., & Yihune, M. (2016). Species composition, relative abundance and habitat association of rodents in Yekoche Forest, East Gojjam, Ethiopia. *International Journal of Biodiversity and Conservation*, 8, 216–223.
- Aplin, K. P., Brown, P. R., Jacob, J., Krebs, C. J., & Singleton, G. R. (2003). Field methods for rodent studies in Asia and the Indo-Pacific. *Acian Monograph No. 100*, 223 pp.
- Asefa, S. (2018). *Diversity, distribution and habitat association of small mammals in Ene* (pp. 1–49) MSc thesis, Addis Ababa University, Addis Ababa.
- Auffray, J., Renaud, S., & Claudae, J. (2009). Rodent biodiversity, human health and pest control in a changing environments. *Kasetsart Journal-Natural Science*, 43, 83–90.
- Bantihun, G., & Bekele, A. (2015). Diversity and habitat association of small mammals in Aridtsy Forest, Awi Zone, Ethiopia. *Zoological Research*, 36, 88–94.
- Bekele, A. (1996a). Rodents of the Menagesha State Forest, Ethiopia, with an emphasis on the endemic *Praomys albipes*. *Tropical Zoology*, 9, 201–212.
- Bekele, A. (1996b). Population dynamics of the Ethiopian endemic rodent, *Praomys albipes* in the Menagesha State Forest. *Journal of Zoology*, 238, 1–12.
- Bekele, A., & Yalden, D. W. (2013). *The Mammals of Ethiopia and Eritrea* (391 pp.). Addis Ababa University Press.
- Bryja, J., Kostinc, D., Yonas, M., Šumberaf, R., Bryjová, A., Kasso, M., Mikula, J., & Lavrenchenko, L. A. (2018). Reticulate Pleistocene evolution of Ethiopian rodent along remarkable altitudinal gradient. *Molecular Phylogenetics and Evolution*, 118, 75–87.
- Caro, T. M. (2001). Species richness and abundance of small mammals inside and outside an African National Park. *Biological Conservation*, 98, 251–257. [https://doi.org/10.1016/S0006-3207\(00\)00105-1](https://doi.org/10.1016/S0006-3207(00)00105-1)
- Chekol, T., Bekele, A., & Balakrishnan, M. (2012). Population density and habitat association of rodents in Pawe Area, Northwestern Ethiopia. *Tropical Ecology*, 53, 15–24.
- Craig, E. W., Bryaj, J., Kerbis, J., & Yonas, M. (2020). Small terrestrial mammal distributions in Simien Mountains National Park, Ethiopia. *Journal of Mammalogy*, 101, 634–647.
- Datiko, D., & Bekele, A. (2013). Species composition and abundance of small mammals in Chebera-Churchura National Park, Ethiopia. *Journal of Ecology and the Natural Environment*, 5, 95–102. <https://doi.org/10.5897/JENE12.045>
- Datiko, D., Bekele, A., & Belay, G. (2007). Species composition, distribution and habitat association of rodents from Arbaminch forest and farmlands, Ethiopia. *African Journal of Ecology*, 45, 651–657. <https://doi.org/10.1111/j.1365-2028.2007.00789.x>
- Dinaw, T., Defar, B., & Genzebe, H. (2017). Species abundance, structure and seasonal variation of small mammals in the CCNP. *African Journal of Ecology and Ecosystem*, 4, 234–241.
- Dubale, M., & Ejigu, D. (2015). Diversity, relative abundance and habitat association of rodents in Aquatimo forest patches and adjacent farmland, East Gojjam, Ethiopia. *Ethiopian Journal of Biological Science*, 14, 201–216.
- Esubalew, E., Giday, K., Hishe, H., & Goshu, G. (2019). Carbon stock of woody species along Altitude gradient in Alemsaga Priority State Forest, South Gondar, NorthWestern Ethiopia. *International Journal of Environmental and Agriculture, Research*, 5, 13–21.
- Fekadu, A., Bekelem, A., & Datiko, D. (2015). Comparative study of species composition, relative abundance and distribution of rodents between exclosure and control sites in the Web Valley of the Bale Mountains National Park, Ethiopia. *Journal of Zoology*, 30, 57–64.
- Geleta, M. (2010). *Species richness, abundance and habitat preference of rodents in Komto Protected Forest, Western Ethiopia* (pp. 1–98). MSc thesis, Addis, Ababa University.
- Getachew, G., Balakrishnan, M., & Bekele, A. (2016). Species composition and habitat association of rodents in Yetere Forest, Central Ethiopia. *International Journal of Ecology and Environmental Science*, 42, 193–200.
- Girma, Z., Bekele, A., & Hemson, G. (2012). Small Mammals of Kaka and Hunkolo, Southeast Ethiopia. *Tropical Ecology*, 53, 33–41.
- Habtam, T., & Bekele, A. (2008). Habitat Association of Insectivores and Rodents of Alatis National Park, Northwestern Ethiopia. *Tropical Ecology*, 49, 1–11.
- Habtam, T., & Bekele, A. (2012). Species composition, relative abundance and habitat association of small mammals along the altitudinal gradient of Jiren Mountain, Jimma, Ethiopia. *African Journal of Ecology*, 51, 37–46. <https://doi.org/10.1111/aje.12005>
- Happold, D. C. D. (2013). *Mammals of Africa. Vol. III: Rodents, hares and rabbits* (84 p). Bloomsbury Publishing.
- Kassa, D., & Bekele, A. (2008). Species composition, abundance, distribution and habitat association of rodents of Wondo Genet, Ethiopia. *Ethiop. Journal of Science*, 31, 141–146.
- Kasso, M., & Bekele, A. (2011). Seasonal population dynamics of rodents of Mount Chilalo, Arsi, Ethiopia. *Ethiopian Journal of Science*, 34, 123–132.
- Kasso, M., & Bekele, A. (2017). Diversity, abundance and distribution of mammals in fragmented remnant forests around Asella Town, Ethiopia. *Journal of Biology*, 1, 1–12.
- Kasso, M., Bekele, A., & Graham, H. (2010). Species composition, abundance and habitat association of rodents and insectivores from Chilalo-Galama Mountain Range, Arsi, Ethiopia. *African Journal of Ecology*, 48, 1105–1114.
- Kingdon, J. D., Happold, T., Butynski, M., Hoffmann, M., & Happold, J. K. (2013). *Mammals of Africa*, Vol. 6. Bloomsbury Publishing.
- Kostin, D. K., Aleksey, A. M., Valeria, A. K., Dmitriy, A. Y., Yihune, M., Kasso, M., Bryja, J., & Lavrenchenko, L. A. (2020). Rodents of Choke

- Mountain and surrounding areas thiofia, the Blue Nile gorge as a strong biogeographic barrier. *Journal of Vertebrate*, 69, 1–12.
- Lavrenchenko, L. A., & Bekele, A. (2017). Diversity and conservation of Ethiopian mammals: What have we learned in 30 years? *Ethiopian Journal of Biological Science*, 16, 1–20.
- Lavrenchenko, L. A., Verheyen, E., & Potapov, S. G. (2004). Divergent and reticulate processes in evolution of Ethiopian *Lophuromys flavopunctatus* species: From mitochondrial and nuclear DNA differentiation patterns. *Journal of the Linnean Society*, 83, 301–316.
- Mahlaba, T. A. M., & Perrin, M. R. (2003). Population dynamics of small mammals at Mlawula, Swaziland. *African Journal of Ecology*, 41, 317–323. <https://doi.org/10.1111/j.1365-2028.2003.00476.x>
- Moges, E., Bekele, A., & Balakrishnan, M. (2016). Species composition, habitat association and relative abundance of small mammals in Borena Sayint National Park, South Wollo, Ethiopia. *Journal of Science*, 39, 50–58.
- ORDA (Organization for Rehabilitation and Development of Amhara) (2010). *Wild life and Natural habitat Assessment in South Gondar Administrative Zone*.
- ORDA (Organization for Rehabilitation and Development of Amhara) (2012). *Participatory Alesaga Priority State Forest resources assessment*. Center for International Forestry Research.
- Shannon, C. E., & Weaver, W. (1949). *A mathematical theory of communication*. (pp. 117). University of Illinois Press.
- Simeneh, G. (2016). *Species composition, distribution, abundance and economic importance of rodents in Choke Mountains, East Gojjam Zone* (pp. 1–149). PhD dissertation, Addis Ababa University, Addis Ababa.
- Vaughan, J. A., Ryan, J. M., & Czaplewski, N. J. (2000). *Mammalogy* (4th ed., 565 pp.). Saunders College Publishing.
- Vieira, M. V. (2003). Seasonal niche dynamics in coexisting rodents of the Brazilian Cerrado. *Studies on Neotropical Fauna and Environment*, 38, 7–15. <https://doi.org/10.1076/snfe.38.1.7.14034>
- Workineh, S., Bekele, A., & Balakrishnan, M. (2011). species diversity and abundance of small mammals in Nechisar National Park, Ethiopia. *African Journal of Ecology*, 50, 102–108.
- Workineh, S., & Reddy, R. U. (2016). Species composition and habitat association of mammals in Nechi Sar National Park, Ethiop. *Journal of Sciences*, 26, 62–71.
- Yalden, D. W., & Largen, M. J. (1992). The mammals of Ethiopia. *Mammalogy*, 22, 115–150.
- Yihune, M., & Bekele, A. (2012). Diversity, distribution and abundance of rodent community in the afro-alpine habitats of the Simien Mountain National Park, Ethiopia. *International Journal of Zoological Research*, 8, 137–149.

How to cite this article: Takele, B., Yihune, M., & Bekele, A. (2022). Composition, abundance and distribution of rodent species in Alesaga Priority State Forest and farmlands, northwestern Ethiopia. *African Journal of Ecology*, 60, 447–455. <https://doi.org/10.1111/aje.12976>